

AI Agents system for Trip Planning

Planning a group trip is often overwhelming—different people, different preferences, endless back-and-forths, and too many open tabs. This project empowers agents each with specialized roles and capabilities to work together using a multi agent system. From comparing destinations to building detailed itineraries and even pulling in data from external systems, you can create these ai agents to coordinate intelligently and simplify decision-making. It's like having a system of expert trip planners working just for you—smart, efficient, and built for how people travel together.

Technical Requirements

- CrewAI framework for orchestrating multi-agent collaboration
- OpenAI API (GPT-4/GPT-3.5) for conversational reasoning and planning
- Serper.dev or similar for real-time search capabilities
- Browseless / LangChain BrowserTools for web scraping and itinerary research
- Ollama (optional) for integrating local language models
- Geolocation and weather APIs for contextual decision-making
- Calendar API (Google Calendar) for itinerary scheduling
- Flight and hotel APIs (Skyscanner, Booking.com) for availability and price checks

Data Sources

- City comparison data from travel APIs like Skyscanner, TripAdvisor.
- Local insights and cultural tips from public sources like Reddit, and You Tube travel vlogs
- Scheduling patterns from user inputs and historical itinerary datasets
- Cost-of-living and price data from Booking.com, and CurrencyLayer APIs
- Bookable options and availability from APIs like Expedia, and Google Travel
- Weather and seasonality trends from OpenWeatherMap and Visual Crossing
- Geolocation and navigation data from Google Maps and OpenStreetMap
- Calendar syncing and event scheduling through Google Calendar API

Solution Approach

The first step to building this project is to define each agent's function using CrewAI framework: Destination Data Researcher Agent to compare multiple cities using web tools and search APIs. Local Expert Agent to provide insider knowledge, cultural tips, and must-visit spots for each city. Itinerary Planner Agent to design a customized day-by-day schedule based on preferences. Budget Advisor Agent to provide cost estimates for travel, stay, food, and experiences. Booking Assistant Agent to interface with external systems to find and suggest bookable options.

External tools like Google Calendar, weather APIs, and geolocation services are embedded into the agent toolkits for context-aware recommendations. The system also supports various deployment modes—whether through GPT-4, GPT-3.5, or local models powered by Ollama for flexibility in performance and cost. The final output can be visualized as a clean, exportable itinerary or even synced directly to the user's calendar, making the entire trip-planning experience effortless and AI-driven.